

Mayank Kundalwal

Curriculum Vitae

Department of CSE
Indian Institute of Technology Jodhpur
☎ (+91) 9261444448
✉ kundalwal.1@iitj.ac.in
🌐 My Webpage
🌐 LinkedIn 🎓 Google Scholar



Research Summary

My research focuses on federated learning and representation learning for privacy-sensitive and heterogeneous environments. I develop robust and interpretable models for medical imaging and distributed systems. My work includes publications in IEEE Transactions and WACV, along with contributions to defense-funded radar signal processing projects.

Education

- 2019–2025 **PhD, Computer Science & Engineering, Indian Institute of Technology, Jodhpur,**
Towards Robust Privacy-preserving Federated Learning under Heterogeneous Environments for Medical Imaging Applications.
- 2016–2018 **M.Tech, Computer Science & Engineering, NIT Patna,**
Privacy-Preservation Framework for Cloud-Based Healthcare Systems CGPA: 8.24.
- 2012–2016 **B.Tech, Computer Science & Engineering, Poornima University,** CGPA: 7.38.

Research Interests

Federated Learning, Computer Vision, Representation Learning, Medical Image Analysis

Teaching Experience

- Jan 2026–Present **Assistant Professor, Manipal University Jaipur.**
Teaching undergraduate courses, conducting lectures and tutorials, mentoring students, and designing assessments. Focus on building strong conceptual understanding through interactive and board-oriented teaching approaches.

Technical Skills

Programming Languages: Python (advanced), C/C++, Java
Deep Learning Frameworks: PyTorch, TensorFlow, Keras
Domain Expertise: Federated Learning, Computer Vision, Representation Learning

Research Projects

- 2023–Present **Confidential Defense Project (Indian Army), Researcher.**
Developing long-range moving target detection algorithms using pulse-doppler radar. The project aims to enhance object identification accuracy at distances of approximately 20 km under challenging environmental conditions. Responsibilities include radar signal processing, Doppler-based motion analysis, and algorithm prototyping for detection and tracking.
(Certain details are classified and cannot be disclosed publicly.)

Industry Experience

- 2018–2019 **Cloud Engineer, Cloudminister Technologies Pvt. Ltd., Jaipur, India.**
Worked on cloud infrastructure and distributed systems, including server management, virtualization, and deployment. Gained experience in maintaining scalable and reliable cloud-based services and supporting system operations.

Publications

Journal Articles

- 2025 Mayank Kumar Kundalwal and Deepak Mishra. Flame: Federated learning with masked autoencoders and mean-prototypes embedding for sparsely labeled medical images. *IEEE Transactions on Emerging Topics in Computational Intelligence*. IEEE, 2025.
- 2025 Mayank Kumar Kundalwal and Deepak Mishra. Ar2fl: Anomaly-resistant robust framework for federated learning. *IEEE Transactions on Artificial Intelligence*. IEEE, 2025.
- 2019 Mayank Kumar Kundalwal, Kakali Chatterjee, and Ashish Singh. An improved privacy preservation technique in health-cloud. *ICT Express*, volume 5, pages 167–172. Elsevier, 2019.

In Conference Proceedings

- 2026 Mayank Kumar Kundalwal, and Deepak Mishra, Federated Model Synchronization for Diagnostic Redefinition through a Novel Selective Parameter Unlearning, In 2026 *IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, IEEE, 2026.
- 2024 Mayank Kumar Kundalwal, Anurag Saraswat, Ishan Mishra, and Deepak Mishra. Client contribution normalization for enhanced federated learning. In 2024 *IEEE 21st India Council International Conference (INDICON)*, pages 1–6. IEEE, 2024.
- 2022 Mayank Kumar Kundalwal, Anurag Saraswat, Ishan Mishra, and Deepak Mishra. BafI: Federated learning with base ablation for cost effective communication. In 2022 *26th International Conference on Pattern Recognition (ICPR)*, pages 1922–1928. IEEE, 2022.
- 2018 Mayank Kumar Kundalwal, Ashish Singh, and Kakali Chatterjee. A privacy framework in cloud computing for healthcare data. In 2018 *International conference on advances in computing, communication control and networking (ICACCCN)*, pages 58–63. IEEE, 2018.

Submitted / Under Review Publications

- 2025 Mayank Kumar Kundalwal, and Deepak Mishra, Context-Aware Hierarchical Swarm Learning for Medical Image Applications, In *ACM Transactions on Multimedia Computing, Communications, and Applications (TOMM)* (Under Review).
- 2026 Mayank Kumar Kundalwal, and Deepak Mishra, Selective Federated Unlearning through Anchor Editing in Embedding Space, In *IEEE Internet of Things Journal*(Under Review).

Achievements

- 2026 **DEI Travel Grant Recipient** – IEEE/CVF (WACV 2026).
- 2025 **Poster Presentation:** “Graduate Forum” at *IndoML 2025*, BITS Pilani – Hyderabad Campus
- 2023–Present **Cyber Volunteer Squad Member** – Ministry of Home Affairs, Government of India. Recognized at the *G20 Conference on Crime and Security in the Age of NFTs, AI, and the Metaverse* (2023).
- 2023–2025 **Indian Army (Defense R&D) Research Funding** – Radar-based Moving Target Detection Project.
- 2019–2023 **Ministry of Education (MoE) Doctoral Fellowship** for PhD Research.
- 2016–2018 **MHRD Fellowship** for M.Tech Studies.
- Reviewer for leading journals (*IEEE TAI, TETCI, IoT Journal, TDSC*) and top conferences (*WACV, ICPR, IJCAI, ECAI*).

Referees

Dr. Deepak Mishra

Assistant Professor, Dept. of CSE
Indian Institute of Technology, Jodhpur
✉ dmishra@iitj.ac.in

Dr. Suman Kundu

Assistant Professor, Dept. of CSE
Indian Institute of Technology, Jodhpur
✉ suman@iitj.ac.in